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Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis	Improving the Reliability and Accuracy of Dialogue-Based Natural Language Generation		
Date of Submission	2026-06-04		

Reviewer

Name	Prof. Wei Ke	Date	2026-06-07
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

This doctoral thesis advances dialogue-based natural language generation by tackling role sensitivity, topic disentanglement, dynamic semantic adaptation, and reliability. It introduces Role-Aware Adapters for speaker-specific extraction, HiSum for hierarchical topic-driven summarization, DABART for adaptive semantic optimization, and SSCM for hallucination mitigation in financial text. A systematic evaluation of reasoning-oriented LLMs clarifies their limits in summarization. Collectively, these frameworks improve accuracy, coherence, and faithfulness across customer service, medical, meeting, and financial domains. A key weakness is reliance on benchmark datasets and limited real-world deployment, leaving practical robustness and scalability underexplored.

Knowledge Base and Literature Review.

The literature review in this thesis is thorough and well-structured, tracing dialogue summarization across generative paradigms, structural modeling, and reliability research. It clearly distinguishes open-domain versus task-oriented summarization, and explains role-oriented and query-focused tasks with strong theoretical grounding. Strength lies in its synthesis of pretrained architectures, PEFT strategies, and reliability concerns such as hallucination mitigation. However, the review is somewhat descriptive, emphasizing taxonomies and technical families without deeply critiquing their limitations or unresolved debates. It could more sharply evaluate gaps in empirical validation and practical deployment, making the critical perspective less forceful than its breadth suggests.

Originality and Independence.

The originality of the thesis methods lies in their systematic attempt to go beyond flat dialogue summarization and tackle structural and reliability challenges. Role-Aware Adapters (RAA) are original in treating role information as a central weighting mechanism rather than an auxiliary signal, enabling a single model to flexibly generate both overall and role-specific summaries. HiSum advances originality by embedding topic segmentation directly into the summarization pipeline through a variational GMM and hierarchical attention, which is more precise than conventional clustering. DABART introduces dynamic semantic optimization, combining adaptive topic weighting and semantic bridging to capture evolving salience --- an uncommon approach compared with static representations. SSCM is distinctive in reframing hallucination mitigation as a ranking problem, training a critical model to filter unreliable outputs without manual annotation. Critically, while these designs are conceptually novel and technically creative, their originality is somewhat bounded by reliance on benchmark datasets and controlled evaluation settings. Real-world deployment, broader multilingual validation, and integration with continuous experimental feedback remain underexplored, meaning originality is strong in methodological innovation but less proven in practical robustness.

Arguments and Reasoning.

The arguments and reasoning behind the thesis methods are innovative but not without weaknesses. Role-Aware Adapters (RAA) argue that role information should be central to summarization, not auxiliary. This reasoning is strong, as speaker roles clearly shape meaning, but it assumes role prompts and weighting alone can fully capture nuanced intent, which may oversimplify complex interactions. HiSum reasons that latent topic segmentation plus hierarchical attention improves coherence. This is persuasive, since interleaved topics are a real challenge, yet the reliance on fixed clustering parameters may limit adaptability to diverse dialogues. DABART argues that dynamic topic weighting and semantic bridging better reflect evolving salience. The reasoning is compelling, but it risks instability if semantic shifts are subtle or noisy, and its improvements are modest in simpler dialogues. SSCM reasons that hallucination reduction can be reframed as ranking, separating generation from reliability judgment. This is a clever argument, but it depends heavily on the quality of automatically constructed contrastive data, which may introduce bias. Overall, the reasoning across methods is creative and well-aligned with real challenges, but sometimes over-relies on controlled benchmarks and lacks deeper validation in unpredictable, real-world dialogue contexts.

Evaluation within the Field.

This thesis positions itself at the intersection of dialogue summarization and reliability-sensitive generation, advancing beyond conventional extractive or flat seq2seq approaches. Its originality lies in systematically embedding role awareness (RAA), hierarchical topic disentanglement (HiSum), dynamic semantic adaptation (DABART), and hallucination control (SSCM). Compared with the broader field, it contributes a coherent methodological toolkit that directly addresses speaker attribution, topic interleaving, evolving salience, and factual faithfulness --- issues often treated separately in prior work. Critically, while the research strengthens structural and reliability modeling, its position remains somewhat limited by reliance on benchmark datasets and controlled experiments.

Completeness and Coherence.

The research in this thesis can be more complete. It makes strong contributions by introducing role-aware adapters, hierarchical topic segmentation, dynamic semantic optimization, and a self-supervised reliability model. These methods advance dialogue summarization and hallucination mitigation conceptually. However, practical robustness, scalability, and continuous experimental feedback remain underexplored, leaving the research promising but somewhat incomplete.

Conclusion.

The conclusion of this thesis is well-structured and synthesizes the diverse methodological contributions into a coherent narrative. It emphasizes that dialogue-based generation requires more than fluency: role sensitivity, topic disentanglement, dynamic semantic adaptation, and reliability control are positioned as complementary pillars. A strength is the clear progression from RAA to HiSum to DABART, showing how structural modeling evolves, and the extension to SSCM highlights reliability in high-stakes domains. The reasoning-LLM evaluation adds critical boundary analysis, clarifying that explicit reasoning does not necessarily improve summarization.

Critically, the conclusion acknowledges limitations --- benchmark dependence, uneven domain coverage, and lack of unified integration --- demonstrating self-awareness. However, while future directions are outlined, the discussion could more forcefully address scalability, multilingual robustness, and deployment challenges. Overall, the conclusion succeeds in connecting individual methods to broader design principles, but its critical reflection remains somewhat bounded by academic evaluation rather than practical implementation.

Additional Comments.

